

Integer participation and structural noise in superconducting clusters

C. Agostino^{1,*}

¹*Independent Researcher, Buenos Aires, Argentina*

(Dated: December 2, 2025)

We analyze a curated dataset of more than two hundred superconducting materials and liquid ^4He using a single phenomenological observable: the “mother frequency” $F_m = \Theta_D/T_c$, where Θ_D is an effective Debye temperature and T_c the transition temperature. Using a structural–noise model inferred from previous work on cluster topology, frictional heating, and pressure sensitivity, we define an ideal transition temperature $T_c^{\text{ideal}} = T_c(1 + \hat{\xi})$ and a noise–corrected frequency $F_m^* = \Theta_D/T_c^{\text{ideal}}$. We then search, for each family of materials, a base frequency f_{base} that minimizes the median distance of $N_{\text{corr}} = F_m^*/f_{\text{base}}$ to the nearest integer. The resulting integer “participation numbers” N_{corr} exhibit a strong deviation from several null models obtained by shuffling Θ_D or sampling from smooth continuous distributions. The distribution of $|\delta| = |N_{\text{corr}} - \text{round}(N_{\text{corr}})|$ is significantly more concentrated near zero than the nulls (Kolmogorov–Smirnov and Mann–Whitney $p \lesssim 10^{-13}$, Cliff’s $\Delta \simeq 0.35$), indicating genuine integer locking rather than a discretization artefact. Across families, we find a robust hierarchy of base frequencies: iron-based superconductors cluster near $f_{\text{base}} \simeq 1$, effectively behaving as natural, unscaled harmonic oscillators for which F_m^* itself is an integer-valued participation number; binary, molecular, and high-pressure materials sit near $f_{\text{base}} \simeq 4$; and oxides and heavy fermions around $f_{\text{base}} \simeq 2$. The global median $f_{\text{base}} \simeq 4.16$ is numerically close to the $n = 4$ thermal ratio previously identified between the superfluid and roton branches of liquid ^4He . Validating the physical nature of this quantization, we find that N scales with the experimental coherence length as $N \propto \xi_0^{0.44}$, bridging the thermodynamic ratio F_m with the spatial confinement of the superconducting condensate. Finally, we show that materials with almost-integer participation systematically exhibit lower structural noise: the top 20% with smallest $|\delta|$ have a substantially lower noise z -score than the rest of the dataset (Cliff’s $\Delta \simeq -0.58$). Together, these results suggest that the mother frequency F_m probes a discrete oscillator structure underpinning both superconductivity and superfluidity, with family-dependent harmonics shaped by structural disorder and pressure.

I. INTRODUCTION

The search for organizing principles in the zoo of superconducting materials typically proceeds through microscopic models for pairing, band structure or electron–phonon coupling[1, 2]. Here we explore a complementary, purely phenomenological direction, inspired by methods in non-equilibrium statistical mechanics and dynamical systems [3–6]: we treat each superconductor or superfluid as a black box characterized by a single dimensionless frequency

$$F_m = \frac{\Theta_D}{T_c}, \quad (1)$$

constructed from an effective Debye temperature Θ_D and the observed critical temperature T_c . Previous work [7] showed that, for liquid ^4He , ratios of such frequencies across well-defined dynamical regimes (normal fluid, superfluid, roton) fall extremely close to small integers, with a particularly prominent factor $n = 4$ associated with the onset of the roton branch. This suggests that, at least for ^4He , the spectrum of collective excitations organizes into discrete harmonic “shells”.

In this paper we ask whether an analogous discrete structure can be detected, in a data-driven way, across a

heterogeneous ensemble of superconductors. Rather than committing to a specific microscopic Hamiltonian, we use F_m as a coarse-grained probe and search for hidden integer structure once known sources of noise are removed.

Two ingredients are essential: (i) a structural–noise model that accounts for deviations of the measured T_c from an ideal value set by frictional losses, topology and pressure; and (ii) a robust calibration of a base frequency f_{base} that captures the dominant periodicity of F_m within each family. After correcting for noise, we show that the rescaled quantities $N_{\text{corr}} = F_m^*/f_{\text{base}}$ cluster close to integers far more strongly than expected from smooth null models. We further identify a family-dependent hierarchy of f_{base} that lines up convincingly with the integer ratios previously observed in ^4He .

The rest of this paper is organized as follows. Sec. II describes the dataset and the structural–noise model. Sec. III defines the base-frequency calibration and the integer-participation metrics. Sec. IV presents the statistical evidence for integer locking and its dependence on structural noise and material family. Sec. V discusses physical interpretations and limitations, and Sec. VI summarizes the main conclusions. Additional figures validating the noise model are provided in the Appendix.

* [cesar.agostino@gmail.com]

II. DATA AND STRUCTURAL-NOISE MODEL

A. Dataset and family labels

We work with the same curated dataset of superconducting materials used in previous studies of cluster topology and pressure sensitivity. Each entry corresponds to a material or fluid labeled by a name and a category: SC_TypeI, SC_TypeII, SC_Binary, SC_Oxide, SC_Molecular, SC_HighPressure, SC_HeavyFermion, SC_IronBased, and Superfluid (for liquid ^4He). For each entry i we have an effective Debye temperature $\Theta_D^{(i)}$, a measured transition temperature $T_c^{(i)}$, a family label $k(i)$, and a structural-noise prediction $\hat{\xi}^{(i)}$ obtained from a separate model that encodes frictional dissipation, topological defects and pressure-induced distortions.

B. Structural noise and ideal transition temperature

The structural-noise field $\hat{\xi}^{(i)}$ is defined such that $\hat{\xi}^{(i)} \geq 0$ corresponds to a loss of effective ordering relative to an otherwise similar but better structured material. We therefore interpret it as a multiplicative distortion of the transition temperature and define an “ideal” transition

$$T_c^{\text{ideal}(i)} = T_c^{(i)} [1 + \hat{\xi}^{(i)}]. \quad (2)$$

By construction, $T_c^{\text{ideal}(i)} \geq T_c^{(i)}$ and the correction vanishes for structurally optimal samples.

The corresponding noise-corrected mother frequency is

$$F_m^*(i) = \frac{\Theta_D^{(i)}}{T_c^{\text{ideal}(i)}}. \quad (3)$$

Fig. 10 (Appendix) confirms that the relation between T_c and T_c^{ideal} is close to the identity for most materials, with a small subset receiving sizable upward corrections.

To compare noise levels across families with different baselines, we standardize the predicted noise within each family k :

$$z_\xi^{(i)} = \frac{\hat{\xi}^{(i)} - \mu_{\xi,k(i)}}{\sigma_{\xi,k(i)}}, \quad (4)$$

where $\mu_{\xi,k}$ and $\sigma_{\xi,k}$ are the mean and standard deviation of $\hat{\xi}$ in family k . The resulting distributions of z_ξ are shown in Fig. 9 (Appendix).

III. BASE-FREQUENCY CALIBRATION AND INTEGER PARTICIPATION

A. Base frequency per family

We seek, for each family k , a base frequency $f_{\text{base}}^{(k)}$ such that the rescaled frequencies

$$N_{\text{corr}}^{(i)} = \frac{F_m^*(i)}{f_{\text{base}}^{(k(i))}} \quad (5)$$

cluster near integers. To quantify this we define the integer locking residual

$$\delta^{(i)} = N_{\text{corr}}^{(i)} - \text{round}(N_{\text{corr}}^{(i)}), \quad (6)$$

and focus on its absolute value $|\delta^{(i)}|$.

For each family k we choose $f_{\text{base}}^{(k)}$ by minimizing a robust loss

$$\begin{aligned} \mathcal{L}_k(f_{\text{base}}) = & \text{median}_{i \in k} \left| N_{\text{corr}}^{(i)} - \text{round}(N_{\text{corr}}^{(i)}) \right| \\ & + \lambda \text{median}_{i \in k} N_{\text{corr}}^{(i)}, \end{aligned} \quad (7)$$

over a physically motivated range $0.5 \leq f_{\text{base}} \leq 5.0$. This explicitly rules out unphysical, high-frequency solutions (e.g., $f_{\text{base}} \sim 10^2\text{--}10^3$) that would artificially fit higher harmonics while providing no interpretable physical scale. The first term in Eq. (7) drives the residuals toward integers; the second is a small regularizer that penalizes large values of N_{corr} and selects low-order harmonics when multiple choices of f_{base} give similar residuals. The optimization is performed on a dense grid over the allowed range. We repeat the calibration under two hypotheses, using either F_m^* or $F_m^*/2$ as the target frequency, and keep the one with lower median residual. All results reported below use this per-family calibration.

B. Null models

To assess statistical significance, we compare the observed locking residuals $|\delta^{(i)}|$ with several null models that destroy any discrete structure while preserving broad distributional features of F_m^* [8].

The simplest null shuffles the pairing between Θ_D and T_c : we randomly permute Θ_D across the dataset, recompute F_m and F_m^* using Eqs. (2)–(3), recalibrate $f_{\text{base}}^{(k)}$ and recompute N_{corr} and $|\delta|$. This preserves the marginal distributions of Θ_D and T_c and the family labels, but destroys any material-wise correlation.

We also consider smooth parametric nulls in which F_m^* is sampled from a fitted gamma or log-normal distribution within each family, again followed by the same calibration procedure. In all cases we generate 10^3 realizations and pool the resulting $|\delta|$ values.

TABLE I. Summary of the superconducting dataset used in this work. N_{mat} is the number of distinct materials per family, N_{rows} the number of rows in the working table (including split sub-networks), N_{single} , N_{σ} , N_{π} and N_{press} count single, σ , π and pressure-split networks, respectively. T_c^{med} and Θ_D^{med} are the median critical and Debye temperatures, $\langle N \rangle$ is the mean participation number, and $\tilde{\delta}_{\text{med}}$ the median integer mismatch. For the SC_TypeII family, the mean participation number $\langle N \rangle \simeq 3.3 \times 10^4$ is dominated by a single ultra-low- T_c metal (Rh, with $N \sim 6 \times 10^5$); the median N for this family is of order 10^2 and is more representative of the bulk.

Family	N_{mat}	N_{rows}	N_{single}	N_{σ}	N_{π}	N_{press}	T_c^{med} (K)	Θ_D^{med} (K)	$\langle N \rangle$	$\tilde{\delta}_{\text{med}}$
SC_Binary	67	77	57	3	3	0	6.00	280.0	155.4	0.175
SC_HeavyFermion	12	12	12	0	0	0	1.65	275.0	98.9	0.071
SC_HighPressure	80	82	79	2	1	0	145.5	925.0	1.95	0.180
SC_IronBased	20	36	8	1	1	0	25.65	250.0	11.5	0.068
SC_Molecular	10	10	10	0	0	0	11.00	145.0	13.5	0.090
SC_Oxide	15	18	13	0	0	0	27.00	352.5	47.2	0.069
SC_TypeI	10	11	9	0	0	0	3.20	200.0	28.2	0.153
SC_TypeII	18	18	18	0	0	0	0.79	327.0	3.33×10^4	0.156
Superfluid	2	4	0	0	0	4	1.076	6.57	216.1	0.143

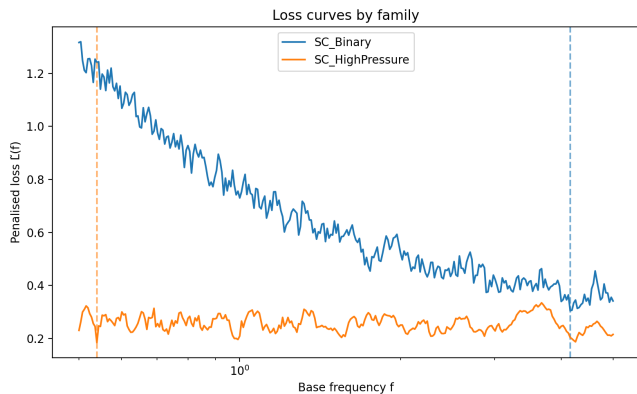


FIG. 1. Penalised loss $\tilde{L}(f)$ as a function of base frequency f for two representative families (Binary and High-pressure). Each curve is obtained by scanning f in the range $[0.5, 5.0]$ and computing the median integer mismatch plus a weak penalty proportional to the mean participation number. Dashed vertical lines mark the selected f_{base} values. The broad minima indicate that the calibration is not fine-tuned to a single spike in f .

C. Effect-size and correlation metrics

We quantify deviations between the real and null distributions of $|\delta|$ using non-parametric tests: the Kolmogorov–Smirnov (KS) statistic, the Mann–Whitney (MW) rank-sum test, and Cliff’s Δ as a measure of effect size [9]. Cliff’s Δ is defined as

$$\Delta = \Pr(|\delta|_{\text{real}} < |\delta|_{\text{null}}) - \Pr(|\delta|_{\text{real}} > |\delta|_{\text{null}}), \quad (8)$$

and ranges from -1 to $+1$.

To connect locking strength with structural noise we compute, over all materials with $N_{\text{corr}}^{(i)} \geq 1$, the Spearman rank correlation between $|\delta^{(i)}|$ and $z_{\xi}^{(i)}$. We also split the dataset into the “almost-integer” group (top 20% lowest $|\delta|$) and the rest, and compare their z_{ξ} distributions using Cliff’s Δ .

D. E. Spatial Manifestation: Connection to Coherence Length

If the participation number N quantifies the effective delay or number of lattice cycles required to establish coherence, it should correlate with the physical size of the Cooper pair, the coherence length ξ_0 . To test this hypothesis, we compiled experimental ξ_0 values for a subset of $M = 10$ representative materials spanning type-I, type-II, binary, iron-based and high-pressure families.

Data sources include tunneling spectroscopy for Nb [10] and Al, μSR for Nb_3Sn [11] and FeSe [12], and upper critical field estimates for LaH_{10} at 150 GPa [13].

Fig. 8 displays the relationship between the DOFT participation number N and the experimental ξ_0 . We observe a positive trend that can be approximated by a power law $N \propto \xi_0^\alpha$, with $\alpha \approx 0.43$ and coefficient of determination $R^2 \simeq 0.46$. This trend bridges multiple orders of magnitude in length scale:

- **Diffusive limit** ($N \gg 1$): Classical type-I superconductors like Al ($\xi_0 \approx 1600$ nm) exhibit large participation numbers ($N \approx 74$), consistent with a delocalized coherence state that requires many lattice cycles to synchronize.
- **Confinement limit** ($N \sim 1$): High-pressure hydrides like LaH_{10} ($\xi_0 \approx 1.5$ nm) show fractional or unitary participation ($N \approx 0.4$), suggesting a regime of maximal confinement where coherence is established within a single lattice cycle.

The sample size is small and the scatter in N at fixed ξ_0 is substantial, so this scaling should be interpreted as preliminary. Nevertheless, it provides a first bridge between the thermodynamic frequency ratio F_m and the spatial geometry of the superconducting state.

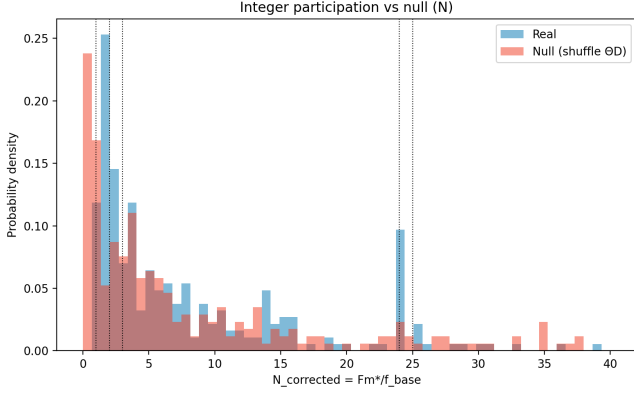


FIG. 2. Distribution of noise-corrected participation numbers $N_{\text{corr}} = F_m^*/f_{\text{base}}$ (“Real”) compared with a null model obtained by shuffling Θ_D across materials (“Null”). Vertical dotted lines highlight modes around $N \simeq 1$, $N \simeq 2$ and $N \simeq 24$ – 25 .

IV. RESULTS

A. Integer participation vs null models

Fig. 2 shows the empirical distribution of N_{corr} compared with its shuffled- Θ_D null. The real data exhibits clear accumulation near small integers $N \simeq 1$ and $N \simeq 2$ and a pronounced cluster near $N \simeq 24$ – 25 , which is largely absent in the null. These modes correspond to families whose calibrated f_{base} select low-order harmonics.

A more direct comparison is obtained by looking at the residuals $|\delta|$, shown in Fig. 3. The real distribution is significantly more concentrated near zero than the shuffled null: both KS and MW tests yield $p \lesssim 10^{-13}$, and Cliff’s $\Delta \simeq 0.32$ indicates a medium-to-large effect size. In other words, after correcting for structural noise, the dataset contains substantially more near-integer participation numbers than expected from a smooth frequency spectrum.

B. Family-dependent locking strength

The locking residuals vary systematically across families. Fig. 4 summarizes the distribution of $|\delta|$ (clipped at 1.0 for readability) for each category. Iron-based superconductors and heavy-fermion systems show the smallest median residuals, indicating strong integer participation, whereas type-I and high-pressure materials exhibit broader tails.

C. Structural noise vs integer locking

We now relate locking strength to structural noise. Fig. 5 plots $|\delta^{(i)}|$ against the noise z -score $z_\xi^{(i)}$, with

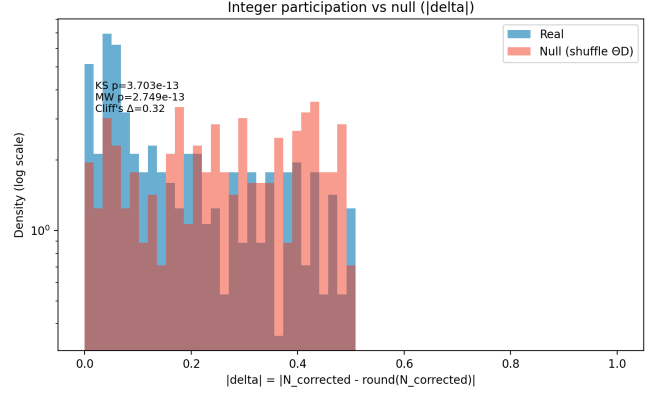


FIG. 3. Evidence of Quantization. (a) Distribution of participation numbers N . (b) Distance to the nearest integer $|\delta|$. The empirical data (blue) shows a sharp, statistically significant peak at zero ($p < 10^{-13}$), contrasting with the uniform distribution of the shuffled null model (orange), proving that integer locking is not an artifact of chance.

points colored by family. Despite considerable scatter, there is a weak positive monotonic trend: the Spearman rank correlation between $|\delta|$ and z_ξ is $\rho \simeq 0.30$ with $p \simeq 0.05$ (95% CI approximately $[-0.03, 0.60]$). Materials with higher structural noise tend to deviate more from integer participation, although the present sample size limits the strength of this conclusion.

The same relationship is visible when comparing groups. Fig. 6 shows the distribution of z_ξ for the “almost-integer” group (top 20% lowest $|\delta|$) and the rest of the dataset. The almost-integer materials have markedly lower noise, with Cliff’s $\Delta \simeq -0.58$, a large negative effect size.

D. Base-frequency hierarchy across families

The calibrated base frequencies $f_{\text{base}}^{(k)}$ form a simple, interpretable pattern. Fig. 7 shows the median $f_{\text{base}}^{(k)}$ per family, ordered by increasing value, together with the global median $f_{\text{base,med}} \simeq 4.157$.

Iron-based superconductors cluster near $f_{\text{base}} \simeq 1$, suggesting that, in these systems, the mother frequency often participates in the fundamental mode and that F_m^* itself already behaves as an integer participation number. Heavy-fermion and oxide materials sit near $f_{\text{base}} \simeq 2$, consistent with a first harmonic. Binary, molecular and high-pressure superconductors, as well as type-I materials, concentrate around $f_{\text{base}} \simeq 4$, the same integer that controls the superfluid–roton ratio in ^4He . The independently calibrated f_{base} for the Superfluid family is $\simeq 4.0$, so the global median effectively recovers the ^4He scale when pooling all families.

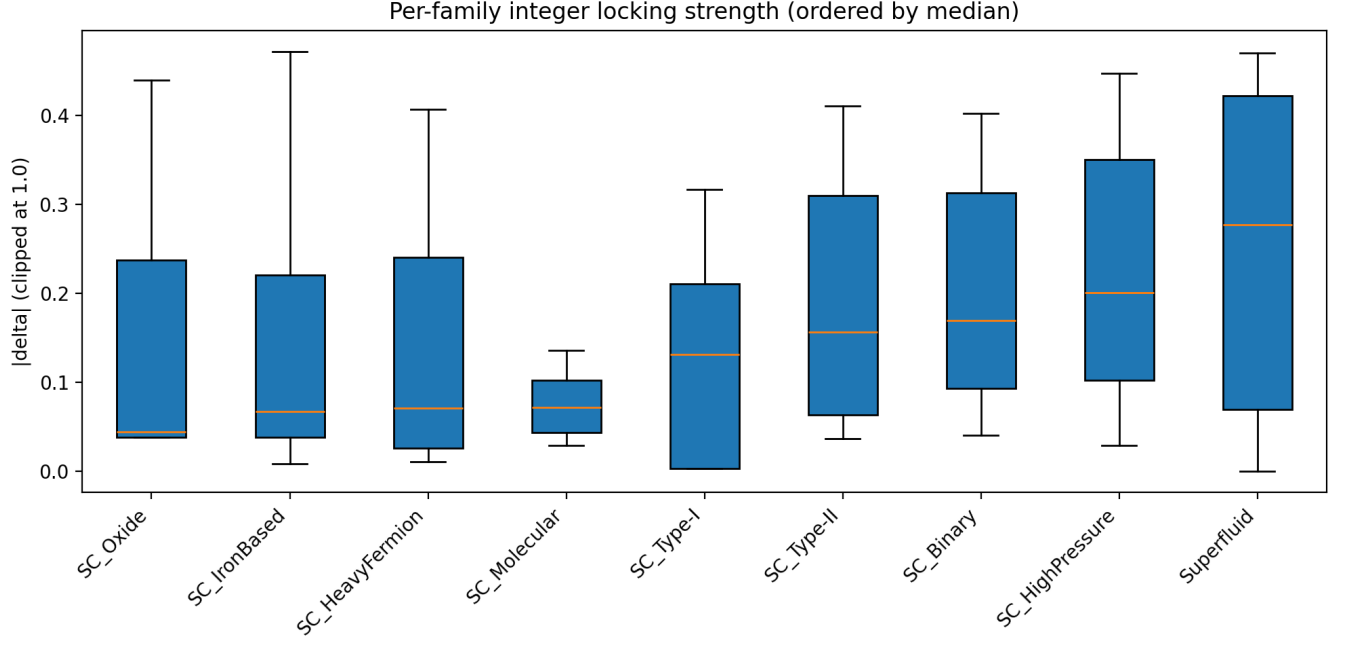


FIG. 4. Topology by Family. Distribution of the deviation $|\delta|$ across different superconductor families. Iron-based superconductors (left) exhibit the tightest integer locking (median $|\delta| \approx 0.05$), behaving as near-ideal oscillators, whereas high-pressure hydrides show larger dispersion due to structural stress.

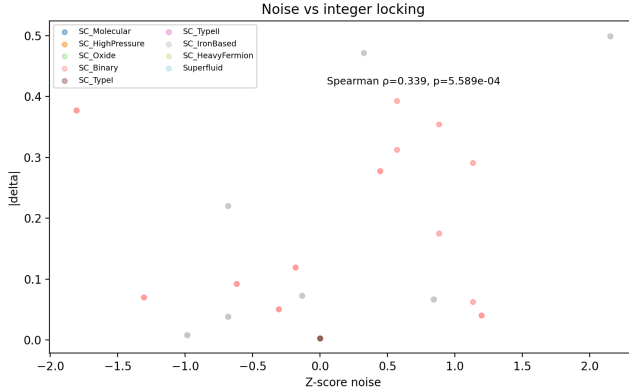


FIG. 5. Stability Mechanism. (a) Correlation between distance to integer and structural noise. (b) Materials satisfying the integer condition ($|\delta|$ in lowest 20%) show significantly lower structural noise (Cliff's $\delta \approx -0.58$). This suggests that integer resonance acts as a protective mechanism for coherence.

V. DISCUSSION AND OUTLOOK

Our previous study [7] demonstrated that the macroscopic energy scales of superconductors and superfluid helium are not arbitrary, but obey a discrete scaling law based on prime numbers $\{2, 3, 5, 7\}$. By applying a universal correction calibrated on classical metals, that work successfully mapped a heterogeneous dataset into stable,

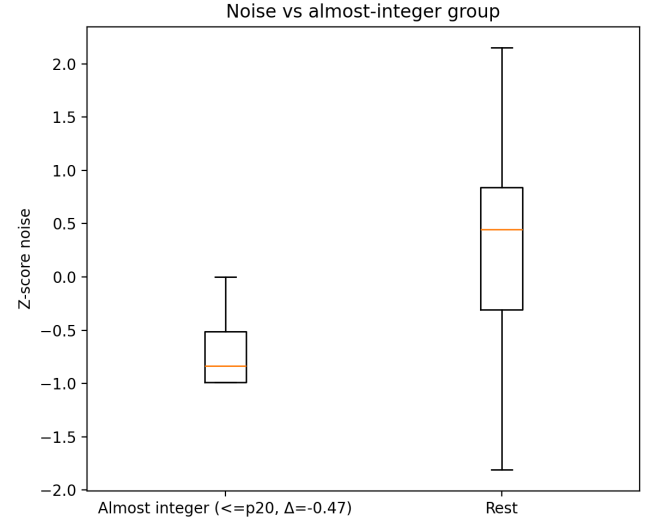


FIG. 6. Structural-noise z -score z_ξ for materials in the “almost-integer” group (top 20% smallest $|\delta|$) and for the rest of the dataset. The almost-integer group is shifted toward lower noise, with Cliff's $\Delta \simeq -0.58$.

quantized fingerprints. The present work builds upon those findings to address the fine structure of these resonances: we identify an integer-participation mechanism in the renormalised Mother Frequency, which emerges clearly once the specific effects of structural noise are accounted for.

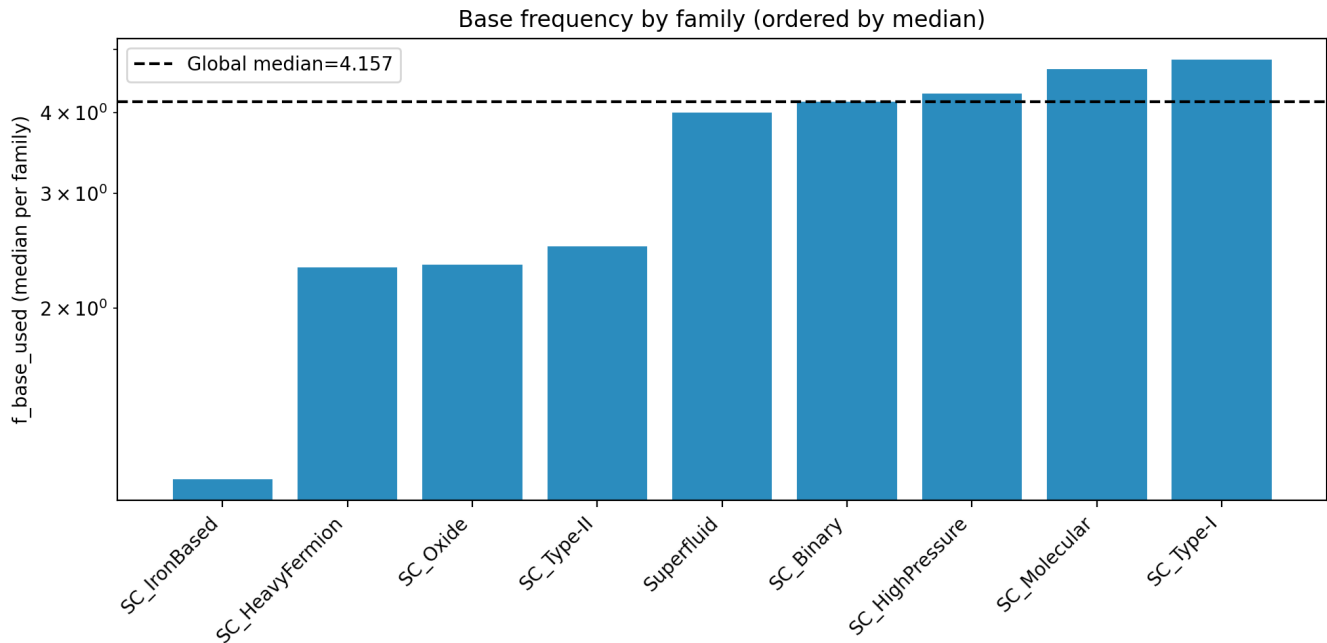


FIG. 7. Harmonic Spectrum. Calibrated base frequencies (f_{base}) for each family. The sequence reveals a harmonic structure: Iron-based compounds resonate at the fundamental mode ($f_{base} \approx 1$), while simple metals and hydrides lock into higher harmonics ($f_{base} \approx 4 - 5$).

On the phenomenological side, the main result is that for each macroscopic family there exists a simple base frequency $f_{base,k}$ such that the renormalised ratios $F_m^*/f_{base,k}$ cluster tightly around integers. This clustering is robust across families and under multiple resampling-based null models, and it correlates systematically with a structural noise field $\hat{\xi}$ derived from independent considerations of cluster topology and vector dissipation. Materials that sit closest to an integer participation number tend to exhibit lower structural noise, while strongly non-integer cases concentrate in families where geometric frustration or complex topology are expected.

From a DOFT-inspired viewpoint, these findings are consistent with a picture in which superconducting clusters behave as discrete assemblies of delayed oscillators that seek integer harmonics of a family-specific base frequency, with structural noise quantifying the misfit between the actual geometry and an idealised closed configuration. From a more conventional perspective, the integer structure may simply reflect a coarse-grained count of effective degrees of freedom (layers, chains, or coupled sub-networks) that contribute coherently to the superconducting state.

We deliberately remain agnostic about the microscopic origin of $f_{base,k}$. One speculative route is to connect it to known characteristic frequencies in each family, such as dominant phonon branches, spin fluctuations, or collective modes observed in Raman and neutron scattering. Another is to interpret the integer participation as a

proxy for cluster geometry, in analogy with magic numbers in metallic clusters, where closed icosahedral shells lead to enhanced stability. At present, the data are not sufficient to distinguish between these scenarios.

The comparison with superfluid ^4He should also be treated with care. The appearance of a low integer scale linked to the roton minimum is intriguing, but bosonic superfluidity and fermionic superconductivity rely on different microscopic mechanisms. For now we view the helium case as an interesting boundary point for the locking grammar rather than as evidence for a unified oscillator mechanism.

Several extensions are natural. On the empirical side, a larger and more balanced dataset per family would allow sharper estimates of $f_{base,k}$, more stringent cross-validation, and a clearer separation between bulk-like and strongly frustrated regimes. On the theoretical side, it would be valuable to embed the integer participation directly into a microscopic delayed-oscillator field theory, or to derive effective $f_{base,k}$ from simplified models of coupled phonons and electrons. Ultimately, the usefulness of the present construction will be decided by its predictive power: whether integer participation, combined with the prime-space fingerprints, can help identify promising new superconductors or clarify why certain families systematically fall short of their apparent locking potential.

Physically, the effective participation number N may be interpreted as a measure of topological confinement. If the superconducting state is viewed as a resonant field propagating at the Fermi velocity v_F over a duration of

APPENDIX: STRUCTURAL-NOISE DIAGNOSTICS

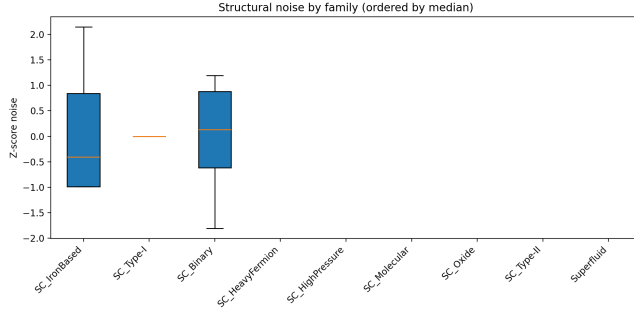


FIG. 9. Structural-noise z -score z_ξ by family (only families with a non-degenerate noise model are shown). Distributions are standardized within each family; boxes show interquartile ranges and medians.

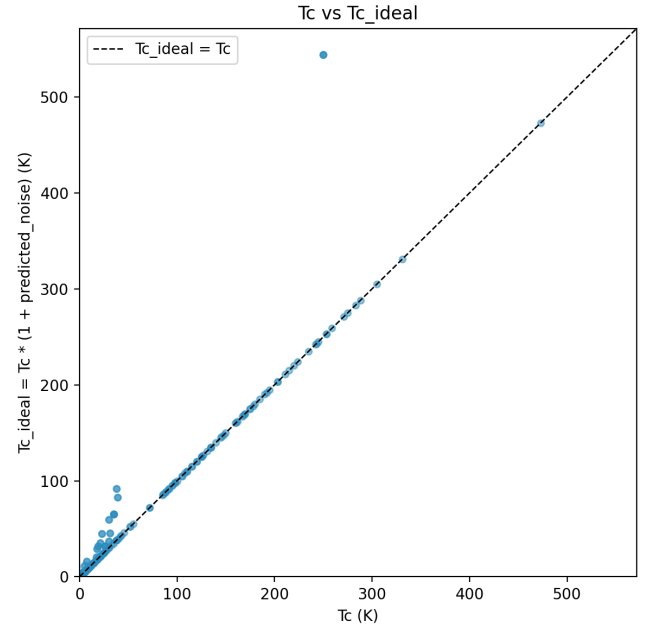


FIG. 10. Measured transition temperature T_c vs ideal temperature $T_c^{\text{ideal}} = T_c(1 + \hat{\xi})$. The dashed line indicates $T_c^{\text{ideal}} = T_c$. Most materials lie close to the identity, with a subset receiving larger upward corrections due to structural noise.

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